Who Gets an ESG Score? Coverage Selection and Firm Value in Five Southeast Asian Markets

Abstract

Most listed firms in Southeast Asia have no environmental, social, and governance (ESG) score, so investors who screen on scores cannot hold them. We ask which firms enter the London Stock Exchange Group (LSEG) ESG database and what happens to their valuation around entry. We compare 675 non-financial firms in Indonesia, Malaysia, the Philippines, Singapore, and Thailand that received a first recorded score with 2,613 never-scored firms. Entry is selective. Scored firms are larger, and on average their market-to-book ratio had grown 10.0% more than that of similar firms in the years before their first recorded score, a run-up not detected in a broad Malaysian wave. After entry, the valuation gap stops widening, but whether coverage itself changed valuations depends on when investors first saw the score and on untestable trend assumptions. Research and investment screens that compare rated with unrated firms should model who gets rated and when.

Keywords: ESG ratings, firm valuation, rating coverage, selection, Southeast Asia, sustainable finance

JEL Classification: G14, G15, G32, M14, Q56

1. Introduction

Environmental, social, and governance (ESG) scores work as an entry ticket: a fund, index, or mandate that may hold only scored stocks cannot buy a firm that no provider covers. In five Southeast Asian equity markets, most listed firms had no score from the London Stock Exchange Group (LSEG) at any point between 2014 and 2024, and much of the academic evidence on ESG ratings uses LSEG data (Berg, Kölbel, et al., 2022; Dobrick et al., 2023). This paper asks which firms enter the scored universe and what happens to their market valuation around entry.

If a score makes a firm visible to more investors, its price should rise, because a wider investor base shares its risk (Merton, 1987); if the score also tells investors whether the firm is green or brown, prices should move in opposite directions for the two groups (Pástor et al., 2021; Pedersen et al., 2021). A provider, however, is likely to add a firm once it is large, liquid, and discloses enough to be scored, and its scores rise with firm size (Dobrick et al., 2023; Drempetic et al., 2020). If firms enter after a period of rising value, a comparison of scored and unscored firms credits the score with growth that came before it. Studies, mostly of U.S. firms, link coverage by more raters to fewer ESG violations (Tsang et al., 2024), lower emissions and more ownership by high-ESG institutions (Bikmetova & Pirinsky, 2026), and dividend increases (Tsang et al., 2025). None examines valuation in emerging markets, where most firms are unscored and coverage has expanded in waves.

We study Indonesia, Malaysia, the Philippines, Singapore, and Thailand. Of the 675 non-financial firms whose first recorded LSEG score falls between 2015 and 2024, 577 had it in 2020 to 2023, and 289 were Malaysian firms first scored in 2021 or 2022. We compare them with 2,613 never-scored firms before and after each firm's first recorded score, the first fiscal year for which the database holds a score, using the estimator of Callaway and Sant'Anna (2021), which avoids the biases of two-way fixed effects regressions under staggered timing (Baker et al., 2022; Goodman-Bacon, 2021). The first recorded score is not necessarily the date on which investors could first see the score (Section 3.3). A pre-analysis plan, committed before the estimation code, records the hypotheses, sample, estimator, and robustness checks; every later analysis is labeled exploratory.

The evidence has three parts. First, entry is selective. Scored firms are larger, and from five years before the first recorded score to the base year (the year before it) their market capitalization grew 14.1% more than that of never-scored firms of similar size, country, and industry (95% interval 3.1% to 25.3%) and their market-to-book (MTB) ratio 10.0% more (1.9% to 18.2%). The run-up is large in the 108 Thai and Singaporean firms first scored in 2019 and 2020 and not detected in the 289 Malaysian firms first scored in 2021 and 2022, the largest wave in the data (95% interval −7.4% to 15.2%). Second, after entry the MTB gap stops widening, but the data do not pin down a causal effect of coverage. Under parallel trends from the base year, the average change in log MTB over the first four years is 0.002 (95% interval −0.046 to 0.048); dated one year later, when a score could first have been public, the change is −6.6%. Most of this difference arises because the later dating measures valuations from the first recorded year, after the cohorts' own rise in that year, and the decline appears only in the waves that had a run-up. Third, a static two-way fixed effects regression on the same data attributes a 6.6% MTB increase to coverage, largely an artifact of comparing firms after entry with the same firms before their growth.

The paper makes three contributions. It documents selection into ESG rating coverage in Southeast Asian markets and adds valuation to the rating-coverage literature (Bikmetova & Pirinsky, 2026; Tsang et al., 2024, 2025). It shows why the valuation effect of coverage is hard to identify: entry follows growth and the recorded first score may precede publication. It also warns that comparisons of scored and unscored firms, and regressions that pool all years around entry, can attribute selection to the score.

2. Related Literature and Hypotheses

2.1 ESG Ratings as Information and Certification

ESG ratings can move prices when they change what investors know or which investors may hold a stock. MSCI downgrades are followed by lower ownership by U.S. funds with ESG mandates and a one-year abnormal return of approximately −2.37%, with weaker effects for upgrades (Berg, Heeb, et al., 2022), and calendar-time portfolios show a similar asymmetry (Shanaev & Ghimire, 2022). Publishing Morningstar sustainability ratings for mutual funds redirected fund flows (Hartzmark & Sussman, 2019).

Certification by inclusion in a sustainability index is the closest analog to a first score. Additions to the Dow Jones Sustainability Index have limited price effects relative to similar firms outside the index (Hawn et al., 2018), and a replication finds no price effect but more analyst following after addition (Durand et al., 2019). In sell-side research, losing analyst coverage lowers prices (Kelly & Ljungqvist, 2012), and the first report on a neglected stock raises its price when the report is favorable (Demiroglu & Ryngaert, 2010).

2.2 ESG Demand, Recognition, and Price Pressure

Two channels predict a valuation response to a first score. Under investor recognition, a stock known to few investors trades at a discount because a small group bears its idiosyncratic risk (Merton, 1987), so recognition predicts a gain for every newly scored firm; S&P 500 additions raise prices permanently, whereas deletions do not lower them permanently (Chen et al., 2004). Under ESG preferences, investors accept lower expected returns on assets they classify as sustainable (Pástor et al., 2021; Pedersen et al., 2021), so a first score predicts gains for firms revealed as green and losses for those revealed as brown, and a pooled average can be close to zero. Both channels work through who trades the stock, and on the Stock Exchange of Thailand local institutional investors adjust their trading to brokerage fees over the long run, whereas foreign and retail investors adjust in the short run (Sudsawasd & Sakdapat, 2026).

Demand shocks also shape realized prices before any rating event. The high realized returns of green assets reflect unexpected increases in ESG concerns rather than high expected returns (Pástor et al., 2022), and part of the price increase at S&P 500 inclusion reverses within weeks (Harris & Gurel, 1986; Shleifer, 1986). A run-up in value before coverage, followed by a flat path, is consistent with a demand shift that ended, with selection on growth, or with both.

2.3 Which Firms Providers Score

We expect providers to cover firms in the indices their clients track and firms that disclose enough to be scored, although our data omit the provider's coverage rules. Mandatory sustainability reporting increases the information available on listed firms (Krueger et al., 2024), and ownership shapes corporate social responsibility performance: among Vietnamese listed consumer-goods firms, outside our sample, foreign ownership is associated with higher corporate social responsibility scores, and state and family ownership with lower ones (Nguyen, 2025).

The provider we study gives higher scores to larger firms (Dobrick et al., 2023; Drempetic et al., 2020) and has rewritten historical scores on a large scale (Berg et al., 2020). Providers disagree about the same firm (Berg, Kölbel, et al., 2022; Christensen et al., 2022), and the disagreement is priced (Avramov et al., 2022; Gibson Brandon et al., 2021). If providers select firms that have grown, a scored-unscored comparison mixes the effect of coverage with the growth that led to it.

2.4 Hypotheses

The pre-analysis plan states three hypotheses, all tested two-sided.

H_1 (primary): The initiation of LSEG ESG coverage changes the log MTB ratio of scored firms relative to comparable never-scored firms. Recognition predicts an increase; preferences predict an increase for high-score firms and a decrease for low-score firms; selection predicts relative growth before coverage and, after it, no effect if the growth was permanent or a decline if it was transient.

H_2 (secondary): Coverage changes log market capitalization, book leverage, and log total assets.

H_3 (heterogeneity): The effect on log MTB differs between firms with initial scores above and below their cohort median. Only the preference channel depends on what the score says, so a difference would point toward it; the absence of a difference cannot rule it out.

Without ownership, fund holding, or trading data, we cannot observe the channels themselves; the data show whether entry follows relative growth, how the valuation gap evolves around entry, and whether that path differs with the initial score.

3. Institutional Setting and Data

3.1 ESG Disclosure and Rating Coverage in the Five Markets

The five markets introduced sustainability reporting rules for listed firms at different times. Bursa Malaysia required a sustainability statement in annual reports for financial years ending on or after December 31, 2016, phased in from the largest companies between 2016 and 2018 (Bursa Malaysia, 2015). Singapore required listed issuers to publish a sustainability report, on a comply-or-explain basis, for financial years ending on or after December 31, 2017 (Singapore Exchange, 2016). Philippine listed companies attached a sustainability report, on a

comply-or-explain basis for the first three years, from the 2019 annual reports (Securities and Exchange Commission Philippines, 2019). Under its original schedule, a 2017 Indonesian regulation applies to listed companies other than small- and medium-asset issuers from 2020 (Otoritas Jasa Keuangan, 2017); the regulator's pandemic relief may have delayed some first reports, a date no variable in our analysis depends on. Thai listed companies have reported sustainability information in the 56-1 One Report since the fiscal year ending in 2021 (Tilleke & Gibbins, 2020).

In Malaysia, FTSE Russell, another part of LSEG, scored only the constituents of the FTSE Bursa Malaysia EMAS index, about 30% of listed companies, until Bursa Malaysia and LSEG agreed in November 2022 to extend these scores to all Main and ACE Market companies (Bernama, 2022). The LSEG ESG score we study, formerly Refinitiv ESG, is a separate product, so our event is the initiation of LSEG ESG coverage, not a firm's first ESG rating of any kind. Coverage arrived in waves: 108 first scores in Thailand and Singapore in 2019 and 2020, and 289 in Malaysia in 2021 and 2022 (Table S1 and Figure S1 of the supplementary material). Of the 675 scored firms, 404 (59.9%) are Malaysian, and Malaysian and Thai firms together make up 84.3%. The waves do not follow a single reporting rule: the Thai wave preceded the One Report requirement, and the Malaysian wave came years after Bursa Malaysia's rule.

3.2 Sample

The data combine the annual LSEG ESG score (LSEG, 2025), not the controversy-adjusted combined score, with accounting and market data from Compustat Global (S&P Global Market Intelligence, 2025) for 3,709 listed firms, each with a row for every year from 2014 to 2024; rows before a firm lists carry no data (3,647 rows). Every firm in the extract has data in 2024, so firms that delisted between 2014 and 2023 are absent from both the scored and the never-scored groups. The sample therefore conditions on survival to 2024, and without delisting data we cannot sign the resulting bias.

Firms that never receive a score in our sample are never-scored and form the comparison group. Dropping financial firms leaves 3,423 firms. A firm's first recorded score year, g, is its first fiscal year with an LSEG ESG score, and its base year is $g - 1$. Only 17 of the 675 scored firms have a later year without a score, so we treat coverage as permanent, the staggered-adoption setting of Callaway and Sant'Anna (2021). Excluding the 135 firms scored in 2014,

which have no pre-coverage year, leaves 3,288 firms with 32,521 listed firm-years: 675 firms first scored between 2015 and 2024 and 2,613 never-scored firms. A firm-year enters an estimate only if its outcome is observed in both the base year and the comparison year (Tables S2 and S3 of the supplementary material). The source data contain no MTB ratio at or below zero, so firm-years with negative book equity appear as missing, like firm-years without market or accounting data. Log MTB ratios are missing for 18.5% of never-scored and 9.0% of scored listed firm-years.

3.3 What the First Recorded Score Dates

The event in this paper is the first recorded score: the first fiscal year for which our extract of the LSEG database holds a score. It may precede the year in which investors first see a score for three reasons. First, the score for fiscal year g draws on reports published after that fiscal year ends, so it reaches investors in year g + 1 or later. Second, the provider has rewritten historical scores on a large scale (Berg et al., 2020), and scores for the five most recent fiscal years can change after publication (Sahin et al., 2023). Third, if the provider backfilled a firm's history at entry, the gap exceeds one year. The first recorded score can therefore date entry too early but, unless the provider deleted an earlier score, not too late.

We could not recover the publication dates. Our extract contains each score by fiscal year, with no publication or update date, and holds the scores in the database at download, after any revision; that every scored firm still has a score in 2024 fits a download of the provider's current universe. LSEG distributes a point-in-time version of its ESG data that keeps values as they were published (LSEG, 2023), but our extract does not contain it. We therefore keep two concepts apart. The first recorded score is what we observe; publicly observable coverage begins at an unknown date from g + 1 onward. We report the valuation path with coverage dated at g, at g + 1, and at g + 2, and draw conclusions only where they hold across these datings. Late dating cannot create the pre-coverage growth we report, because that growth occurs before g, but it can place pre-publication years in the post-coverage window of the baseline estimate.

3.4 Outcomes and Recording Errors

The primary outcome is the natural logarithm of the MTB ratio, the market value of equity divided by its book value. The secondary outcomes are log market capitalization, log total

assets, and book leverage, total debt divided by total assets. Without stock returns or shares outstanding, we use MTB for valuation and treat market capitalization and total assets as measures of scale. Amounts are in U.S. dollars, and we winsorize all four variables at the 1st and 99th percentiles of the pooled sample before taking logarithms.

A few market capitalization records are implausibly large; one Indonesian firm is recorded at 1,138.8 billion U.S. dollars. Market capitalization divided by the MTB ratio gives the implied book equity, which cannot exceed total assets. Of the 27,127 firm-years with all three items, 1,030 have ratios between one and two, which can arise when the items are measured on different dates, and 540 have ratios between two and ten. We flag firm-years in which the implied book equity exceeds ten times total assets, an error of an order of magnitude: 209 firm-years (0.77%), 73 of them for scored firms. A lower threshold, a ratio above two, flags 749 firm-years.

4. Methods

4.1 Target Parameter and Identification

In the notation of Callaway and Sant'Anna (2021), let G_i be the first recorded score year of firm i, with $G_i = \infty$ if it is never scored, and let $Y_{i,t}(g)$ be the potential outcome of firm i in year t if it were first scored in year g. The group-time average treatment effect on the treated (ATT) is

$$ATT(g,t) = E\left[Y_{i,t}(g) - Y_{i,t}(\infty) \vee G_i = g\right] \tag{1}$$

We identify it under conditional parallel trends relative to the base year:

$$E\left[Y_{i,t}(\infty) - Y_{i,g-1}(\infty) \vee X_i, G_i = g\right] = E\left[Y_{i,t}(\infty) - Y_{i,g-1}(\infty) \vee X_i, G_i = \infty\right] \tag{2}$$

where X_i contains country and industry indicators and log total assets and its square, measured in the base year. This assumption concerns changes in outcomes, so level differences between scored and never-scored firms do not violate it, but a different growth path before coverage would.

The pooled sample violates the analogous condition before the base year (Section 5.2). Two arguments make it plausible in parts of the sample, and one argues against it. In the large Malaysian wave the pre-coverage estimates show no detectable run-up, so for these firms the

assumption does not require a trend to stop; the run-up interval is wide, however (−7.4% to 15.2%), and the wave split is exploratory. If the provider adds firms once they cross a size or visibility threshold, relative growth stops being a reason for selection after the threshold, and conditioning on base-year size compares firms on the same side of it. Against it, if part of the run-up was transitory, scored firms would have fallen back even without a score, and the estimates would understate any effect of coverage. We therefore treat the post-coverage estimates as conditional on the parallel trends assumption and give most weight to evidence that does not depend on it: the ordering of growth and entry, and the contrast between waves with and without a run-up.

4.2 Estimation

For each cohort g and year t, we regress the long difference $\Delta Y_{i,t} = Y_{i,t} - Y_{i,g-1}$ on X_i among never-scored firms (Sant'Anna & Zhao, 2020):

$$\Delta Y_{i,t} = X_i' \beta_{g,t} + u_{i,t}, G_i = \infty \tag{3}$$

where $\beta_{g,t}$ is the coefficient vector of the cell and $u_{i,t}$ is the error. We omit the propensity-score model of the doubly robust estimator because total assets overlap poorly: 24.6% of scored firms have larger base-year total assets than the 95th percentile of never-scored firms. Outcome regression does not remove this support problem; it extrapolates the relation estimated among controls to scored firms larger than most controls, so Section 5.6 restricts scored firms to the size range of the controls, in the spirit of the overlap trimming of Crump et al. (2009). The estimated group-time effect is

$$\widehat{ATT}(g,t) = \frac{1}{N_{g,t}} \sum_{i:G_i=g} \left(\Delta Y_{i,t} - X_i' \hat{\beta}_{g,t} \right) \tag{4}$$

where $N_{g,t}$ is the number of firms first scored in year g with data in years g − 1 and t and base-year total assets. We average by event year, e = t − g, weighting cohorts by the number of scored firms (Callaway & Sant'Anna, 2021):

$$\hat{\theta}(e) = \sum_g \frac{N_{g,g+e}}{\sum_h N_{h,h+e}} \widehat{ATT}(g, g+e) \tag{5}$$

where h indexes the cohort. The headline estimate averages, with the same weights, all cells from the first recorded score year to three years later:

$$\hat{\theta}_{post} = \frac{\sum_{(g,t):0 \le t-g \le 3} N_{g,t} \widehat{ATT}(g,t)}{\sum_{(g,t):0 \le t-g \le 3} N_{g,t}} \tag{6}$$

Baseline controls are never-scored firms; later-scored firms join them only in check R1. All comparisons, including pre-coverage ones, are relative to the base year, so a negative $\theta(e)$ for $e \le -2$ means that scored firms grew faster than comparable never-scored firms from year g + e to the base year. Because we measure the covariates in the base year, at the end of any run-up, conditioning on base-year total assets can absorb part of the run-up; Section 5.6 varies this choice.

4.3 Inference

We bootstrap the aggregated effects by resampling firms with their full time series. All specifications share the same 999 draws and fixed seed, so differences between specifications have a valid joint distribution. We report the bootstrap standard error (SE), the percentile 95% confidence interval, a normal-approximation two-sided p-value, and the minimum detectable effect at 80% power in a two-sided 5% test, MDE ≈ 2.8 × SE. We test the primary outcome at the 5% level and Holm-adjust the p-values, but not the intervals, of the three secondary outcomes. A Wald test of the four estimates for e = −5 to −2, with their bootstrap covariance matrix, checks parallel pre-coverage trends; because such tests can have low power and conditioning on them distorts inference (Roth, 2022), we report it only as a diagnostic. Coverage arrived in market-level waves, so a market or industry shock can affect many scored firms at once, as shifts in political uncertainty do for Thai stock prices (Prukumpai et al., 2022); an exploratory check therefore resamples the 50 market-by-industry clusters instead of firms.

4.4 Sensitivity to Departures from Parallel Trends

Because the pre-coverage estimates cast doubt on the parallel trends assumption, we add three exploratory checks. The first removes a linear pre-coverage trend that passes through zero in the base year (row R7); the second adds the growth of log market capitalization from year g − 3 to the base year to X_i (row R8). The third bounds the bias in the spirit of the relative-magnitude

restriction of Rambachan and Roth (2023). Let δ(e) denote the post-coverage violation of the parallel trends assumption in event year e; we assume

$$\left|\delta(e)-\delta(e-1)\right| \le \acute{M} \max_{s \le -1} \left|\hat{\theta}(s)-\hat{\theta}(s-1)\right| \equiv \acute{M}\,\Delta, \delta(-1)=0 \tag{7}$$

where s indexes the pre-coverage event years and Δ is the largest change between consecutive pre-coverage estimates. The bias for event year e is then at most $\bar{M}\Delta(e+1)$, and we widen the headline confidence interval by the weighted average of this bound. With $\bar{M} = 0.25$, the post-coverage gap may drift each year by a quarter of the largest yearly change before coverage.

4.5 Benchmark Regression and Robustness Checks

As a benchmark, we estimate on scored and never-scored firms the static two-way fixed effects (TWFE) regression analyzed by Goodman-Bacon (2021) and Baker et al. (2022):

$$Y_{i,t} = \alpha_i + \lambda_t + \beta D_{i,t} + \varepsilon_{i,t}, D_{i,t} = 1\left[t \ge G_i\right] \tag{8}$$

where α_i and λ_t are firm and year effects, $D_{i,t}$ marks years from the first recorded score onward, and $\varepsilon_{i,t}$ is the error term; standard errors are clustered by firm. The coefficient β compares all post-coverage with all pre-coverage years, so it averages over any run-up and, with staggered timing, uses already scored firms as controls for later cohorts.

The pre-analysis plan fixed six checks, labeled R1 to R6 in the results. They add not-yet-scored firms to the controls (R1), drop the covariates (R2), exclude the 2020 and 2021 cohorts (R3), keep only control firms whose base-year log total assets reach the 10th percentile among scored firms of the same cohort (R4), date coverage three years early as a placebo (R5), and estimate the TWFE regression (R6). All other checks are exploratory.

5. Results

5.1 Which Firms Get Scored

Table 1 compares scored firms in the base year with never-scored firm-years. Scored firms are larger: their median market capitalization is 203.2 million U.S. dollars against 42.2 million, and their median total assets are 4.3 times as large. Their MTB ratio is also higher, while leverage differs little. In Malaysia, where the expansion was broadest, the median total assets of

scored firms, each firm measured by its median over all sample years, were 3.3 times those of the 565 never-scored firms.

Table 1. Characteristics of scored firms in the base year and of never-scored firm-years

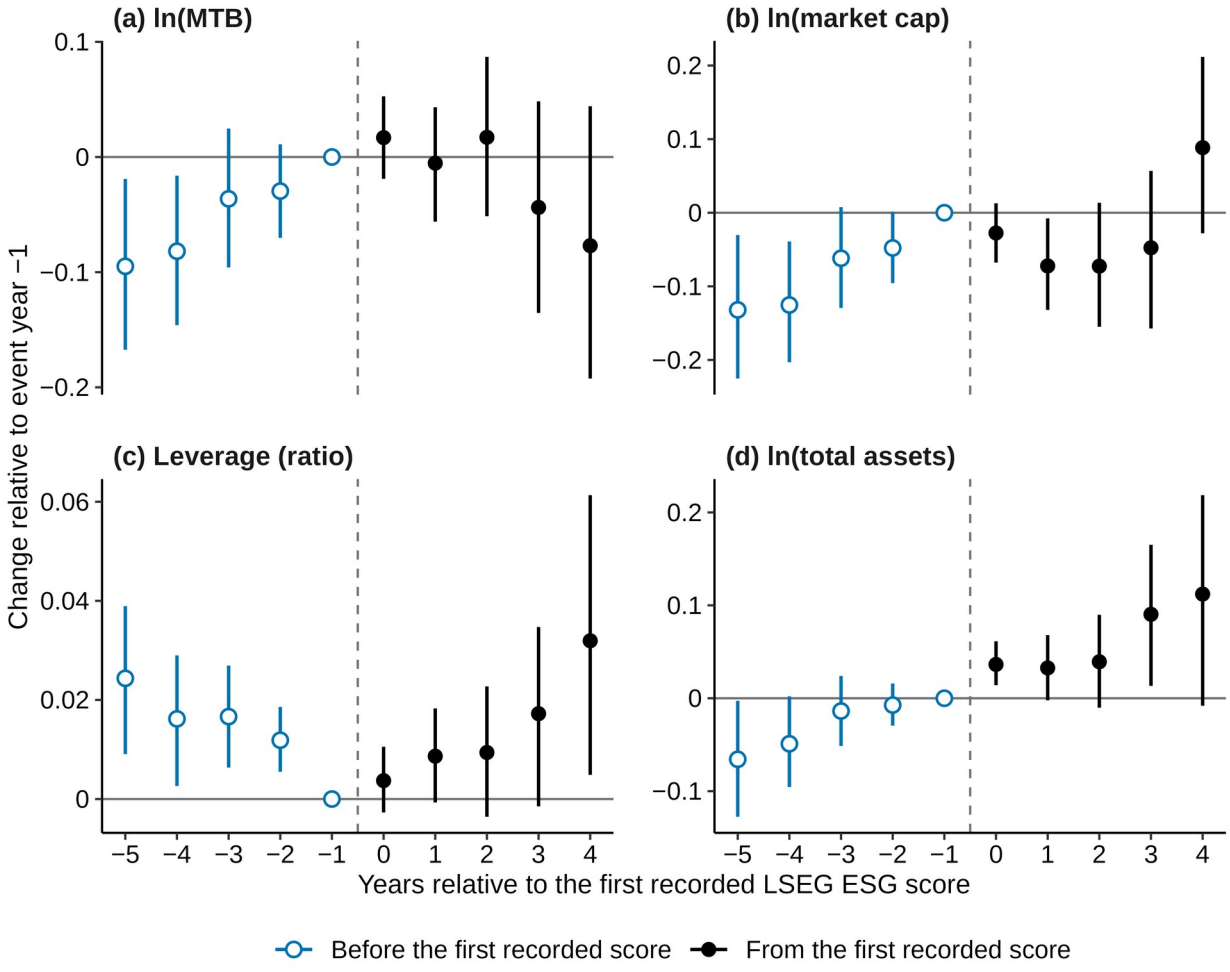
Variable	Scored mean	Scored median	Scored N	Never-scored mean	Never-scored median	Never-scored N	Norm. diff.
Market-to-book ratio	2.37	1.26	629	1.84	0.95	20,706	0.17
Market capitalization (USD million)	762.5	203.2	655	157.3	42.2	22,417	0.69
Book leverage	0.235	0.217	632	0.241	0.205	22,333	−0.04
Total assets (USD million)	882.1	294.3	655	238.0	68.7	24,814	0.61
Return on assets	0.064	0.057	385	0.045	0.046	5,614	0.21

Note: Scored firms in base year g − 1; never-scored firms in 2014–2024. Norm. diff. is the mean difference divided by the root of the average variance.

5.2 The Valuation Path Before and After the First Recorded Score

Figure 1 plots the event-time estimates. Before the first recorded score, the estimates for market capitalization and MTB are negative and rise toward zero; in event years −5 and −4, the MTB estimates are −0.095 and −0.082 ($p = 0.015$ and $p = 0.015$). From event year −5 to the base year, scored firms' market capitalization grew 14.1% more than that of never-scored firms, and their MTB ratio 10.0% more; we call this relative growth the run-up. The joint pre-trend test rejects for market capitalization ($p = 0.022$) but not at the 5% level for MTB ($p = 0.074$). The MTB non-rejection is not evidence of parallel trends: the estimates for event years −5 and −4 differ from zero, and pre-trend tests have low power (Roth, 2022). After the first recorded score, the MTB estimates for event years 0 to 3 lie between −0.044 and 0.017, and none differs from zero at the 5% level; event year 4 rests on only 141 scored firms and is excluded from the headline estimate. The MTB gap thus widens before coverage and stops widening afterward. Only reading the post-coverage part as an effect of coverage depends on the parallel trends assumption.

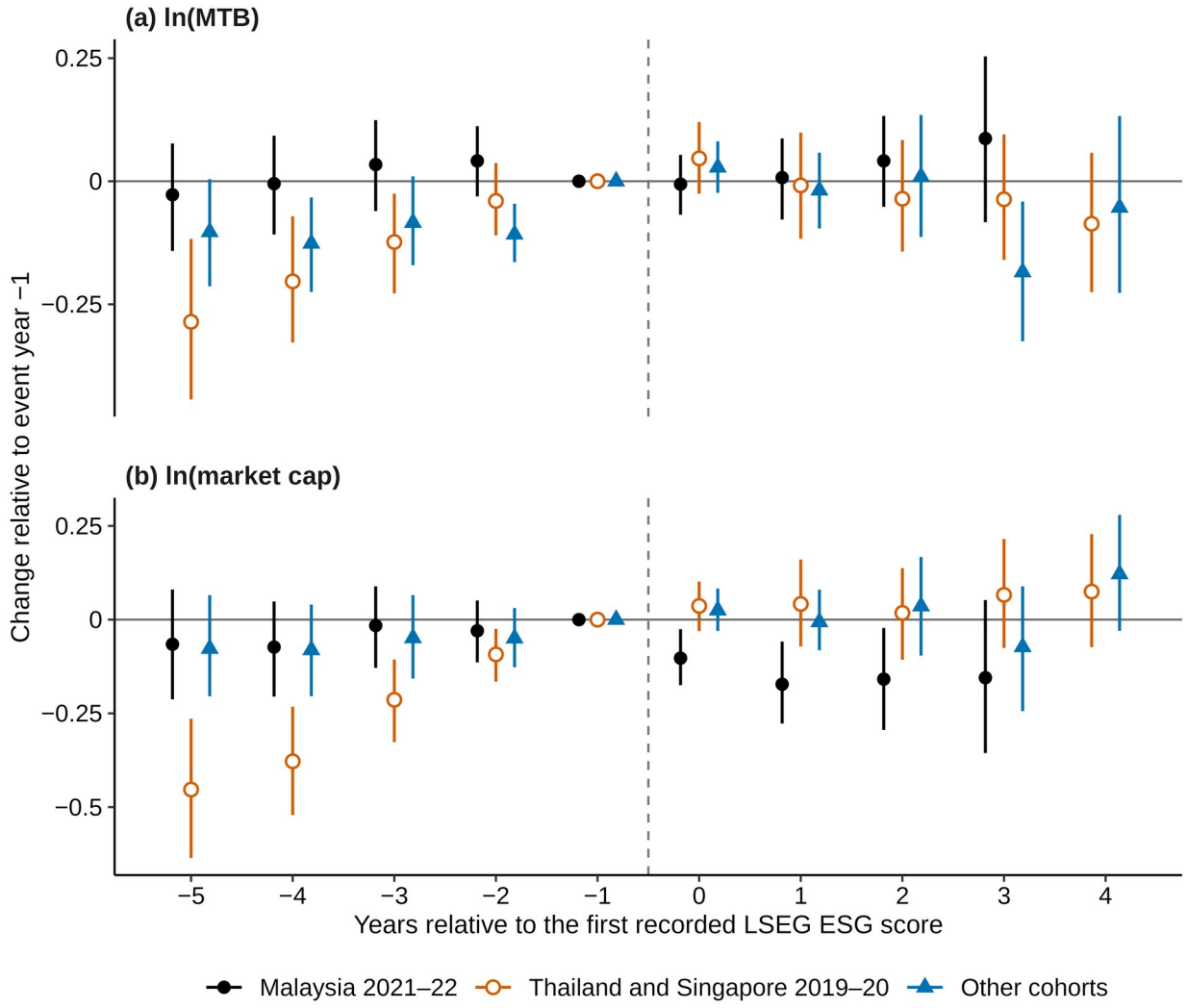
Figure 1. Event-time changes in market-to-book ratio, market capitalization, leverage, and total assets around the first recorded LSEG ESG score



Note: Estimates relative to event year −1 with firm-bootstrap 95% intervals; 141 to 615 scored firms per event year.

Figure 2 separates the two largest coverage waves from the remaining cohorts; because we drew the wave boundaries after seeing the coverage counts, this split is exploratory. The 108 Thai and Singaporean firms first scored in 2019 and 2020 show the largest run-up: 33.1% in MTB relative to never-scored firms (pre-trend p = 0.006). The 289 Malaysian firms first scored in 2021 and 2022 show no detectable run-up, 2.8% (95% interval −7.4% to 15.2%; pre-trend p = 0.549), although a run-up of the pooled size cannot be ruled out, and the 278 firms in the remaining cohorts fall in between. After the first recorded score, no wave shows a detectable MTB gain, while market capitalization fell in the Malaysian wave (−0.146, p = 0.005), which accounts for most of the negative pooled market capitalization estimate.

Figure 2. Event-time changes in market-to-book ratio and market capitalization by coverage wave



Note: Estimates by wave against all never-scored firms, with firm-bootstrap 95% intervals. The waves hold 289, 108, and 278 scored firms.

5.3 Average Change After Entry Under Parallel Trends

Table 2 reports the headline estimates. The change in log MTB is 0.002, with a 95% confidence interval of −0.046 to 0.048, so the data are compatible with changes in the MTB ratio from −4.5% to 4.9%; the MDE is about 7.0%. With the cluster bootstrap, the interval is −0.050 to 0.053. After Holm adjustment, no secondary outcome changes significantly at the 5% level: log market capitalization changes by −0.055 (p = 0.134), leverage by 0.8 percentage points (p = 0.134), and log total assets by 0.044 (p = 0.053). The MTB estimate shows no sign of the gain that the recognition channel predicts (Merton, 1987).

Table 2. Average change after the first recorded LSEG ESG score over event years 0 to 3, under parallel trends

Outcome	ATT	SE	95% CI	p-value	MDE	Pre-trend p	Scored firms	Control firms
ln(MTB)	0.002	0.024	[−0.046, 0.048]	0.936	0.067	0.074	628	2,412
ln(market cap)	−0.055	0.030	[−0.109, 0.003]	0.134	0.084	0.022	635	2,447
Leverage (ratio)	0.008	0.005	[−0.001, 0.018]	0.134	0.014	0.002	631	2,520
ln(assets)	0.044	0.018	[0.007, 0.081]	0.053	0.051	0.083	655	2,603

Note: Outcome regression with firm-bootstrap standard errors; secondary p-values are Holm-adjusted, intervals are not. MDE: 80%-power minimum detectable effect; pre-trend p from the joint Wald test of event years −5 to −2.

Table 3 presents the robustness checks; rows R1 to R6 were pre-specified, and rows R7 to R11 are exploratory. Row R10 dates coverage one year after the first recorded score, the earliest year in which a score could have been public. Its MTB change is −0.068, about −6.6% (95% interval −0.116 to −0.020; p = 0.004), with a joint pre-trend p = 0.072. Pre-specification does not make the recorded year the date on which investors learned of coverage, so we give the two datings equal standing.

Table 3. Robustness checks, timing of coverage, and heterogeneity by initial ESG score

Specification	ln(MTB)	ln(market cap)	Leverage (ratio)	ln(assets)	Scored / control firms
Baseline	0.002 (0.024)	−0.055* (0.030)	0.008* (0.005)	0.044** (0.018)	628 / 2,412
R1 Not-yet-scored in controls	−0.001 (0.024)	−0.053* (0.029)	0.009* (0.005)	0.040** (0.017)	628 / 3,023
R2 No covariates	0.025 (0.022)	0.016 (0.025)	0.012*** (0.004)	0.011 (0.015)	628 / 2,412
R3 No 2020–2021 cohorts	0.031 (0.027)	−0.029 (0.036)	0.004 (0.005)	0.020 (0.019)	413 / 2,407
R4 Size-trimmed controls	0.016 (0.025)	−0.007 (0.031)	0.009* (0.005)	0.101*** (0.017)	628 / 1,975
R5 Placebo, 3 years early	0.059*** (0.023)	0.134*** (0.028)	−0.002 (0.005)	0.115*** (0.015)	540 / 2,032
R6 Two-way fixed effects	0.064** (0.026)	0.185*** (0.032)	0.005 (0.005)	0.117*** (0.022)	3,226 (all)
R7 Pre-trend removed	−0.052 (0.034)	−0.134*** (0.042)	0.023*** (0.006)	0.011 (0.023)	628 / 2,412
R8 Pre-coverage growth	0.028 (0.025)	−0.034 (0.032)	0.008 (0.005)	0.043** (0.018)	567 / 2,156
R9 No Malaysia	−0.003 (0.035)	0.037 (0.038)	0.022*** (0.008)	0.046 (0.029)	256 / 1,926
R10 Dated 1 year later	−0.068*** (0.024)	−0.073** (0.030)	0.007 (0.005)	−0.006 (0.017)	646 / 2,412
R11 Cohorts to 2021	−0.008 (0.033)	−0.030 (0.037)	0.013* (0.007)	0.082*** (0.027)	273 / 2,032
High initial score	−0.019 (0.030)	−0.029 (0.034)	0.009 (0.006)	0.038* (0.022)	312
Low initial score	0.022 (0.033)	−0.081* (0.042)	0.007 (0.006)	0.049** (0.022)	316
High minus low	−0.041 (0.040)	0.052 (0.047)	0.002 (0.008)	−0.011 (0.025)	

Note: Outcome regression with firm-bootstrap standard errors (R6: two-way fixed effects, firm-clustered); firm counts for ln(MTB). *, **, ***: 10%, 5%, 1% (unadjusted).

5.4 Reconciling the Two Datings

Table 4 moves from the first-recorded-year estimate to the one-year-later estimate one step at a time. Row 2 keeps only firms observed in both g − 1 and g, so that every later row draws on the same set of firms; this changes the MTB estimate by −0.005. Row 3 averages event years 1 to 4 instead of 0 to 3; this changes it by −0.016 (p = 0.092). Row 4 moves the reference year from g − 1 to g, which is what dating coverage one year later does to the comparison; this changes the estimate by −0.044 (p = 0.024), 62% of the total gap of −0.070. Measuring size in g (row 5) and returning to all firms (row 6) change little. The steps are sequential, so the shares depend on their order.

Table 4. Reconciling the first-recorded-year and one-year-later datings, step by step

Specification	ln(MTB)	ln(market cap)	Scored firms in averaged event years	Control firms
(1) First recorded year g, reference g − 1, event years 0 to 3, all firms	0.002 (0.024)	−0.055* (0.030)	625	2,412
(2) As (1), firms observed in both g − 1 and g	−0.003 (0.024)	−0.055* (0.030)	615	2,405
(3) As (2), event years 1 to 4	−0.019 (0.030)	−0.053 (0.037)	585	2,251
(4) As (3), reference year g	−0.063*** (0.024)	−0.051* (0.029)	585	2,251
(5) As (4), size measured in g	−0.060** (0.024)	−0.063** (0.031)	583	2,252
(6) One year later, all firms (R10)	−0.068*** (0.024)	−0.073** (0.030)	613	2,407
(7) As (2), cohorts first recorded 2015 to 2020	0.017 (0.040)	0.076* (0.042)	142	1,952
(8) As (3), cohorts first recorded 2015 to 2020	−0.026 (0.047)	0.077 (0.050)	142	1,938
(9) As (4), cohorts first recorded 2015 to 2020	−0.124*** (0.038)	−0.006 (0.040)	142	1,938
(10) As (5), cohorts first recorded 2015 to 2020	−0.121*** (0.038)	−0.018 (0.042)	142	1,938

Note: Outcome regression with firm-bootstrap standard errors; each row changes one feature; firm counts for ln(MTB). *, **, ***: 10%, 5%, 1%.

The reference year thus drives the difference. Moving it from g − 1 to g subtracts from each later estimate the change in log MTB of the same scored firms, relative to the same never-scored firms, between g − 1 and g. The row 4 step, 0.044, is therefore each cohort's own change between g − 1 and g averaged with the weights of event years 1 to 4, which fall most on the early cohorts observed longest after entry; it is an accounting identity, not a separate estimate, and it is larger than the pooled event-year-0 estimate in Figure 1. Part of the decline under the later dating is thus the rise these cohorts had in the first recorded year itself. The cohorts observed through event year 4 (rows 7 to 10) show the same pattern more strongly.

The timing difference is also concentrated in the waves that had a run-up. Under the one-year-later dating, the MTB change is −0.136 for the Thai and Singaporean wave and −0.111 for the remaining cohorts, but 0.012 for the Malaysian wave (Table S6 of the supplementary material). Tested with joint firm-bootstrap draws, the Thai and Singaporean wave differs from the Malaysian wave both in the run-up (difference in the event-year −5 estimate −0.258, $p = 0.010$) and in the one-year-later change (−0.148, $p = 0.006$). The remaining cohorts differ from the Malaysian wave in the one-year-later change (−0.123, $p = 0.010$) but not detectably in the

run-up (p = 0.307). Where no run-up was detected, neither dating detects an MTB change (95% intervals −0.053 to 0.091 and −0.055 to 0.076).

Two readings fit this pattern. If firms were selected at an unusually high valuation that later partly reversed, the one-year-later estimate measures that reversal and could hide a gain from coverage; if valuations would otherwise have stayed at their peak, coverage lowered them. Dating coverage two years later gives −0.048 (95% interval −0.099 to 0.000), and Figure S2 plots the one-year-later path.

5.5 Sensitivity to Departures from Parallel Trends

Row R11 keeps the 7 cohorts observed through event year 3 and gives −0.008 (p = 0.806), so changing cohort composition does not explain the headline estimate. Removing a linear pre-trend (row R7) gives −0.052, and matching on pre-coverage growth (row R8) gives 0.028; both intervals include zero. Under the relative-magnitude bounds (Table S7), with $\bar{M}$ = 0.25 the interval for the MTB effect is −0.071 to 0.073, and any departure with $\bar{M}$ above 0.01 admits a 5% increase. Under the one-year-later dating, the decline loses significance with $\bar{M}$ = 0.25, when its interval runs from −0.138 to 0.002. Departures from parallel trends smaller than those observed before coverage are therefore enough to change the reading of either dating.

5.6 Firm Size and Common Support

Table 5 varies the size adjustment and the support. Across functional forms (none, linear, quadratic, cubic, quintile indicators, or size measured at g − 3), the MTB estimate stays between −0.010 and 0.025. Keeping scored firms no larger than the 95th percentile of controls in their cell retains 487 scored firms and gives −0.021 (p = 0.426); the 90th and 99th percentile cut-offs give −0.013 and −0.006. Within this support the doubly robust estimator of Sant'Anna and Zhao (2020), with control propensity scores capped at 0.99, gives −0.010 (p = 0.715), and only 1.1% of scored firm-cells have an estimated propensity score above 0.95. The MTB result does not depend on extrapolating to the largest firms.

Table 5. Size adjustment and common support

Specification	ln(MTB)	ln(market cap)	Scored firms in averaged event years	Control firms
Baseline: quadratic in log total assets at g − 1	0.002 (0.024)	−0.055* (0.030)	625	2,412
No covariates	0.025 (0.022)	0.016 (0.025)	625	2,412
Linear in log total assets	−0.003 (0.023)	−0.033 (0.029)	625	2,412
Cubic in log total assets	−0.009 (0.024)	−0.055* (0.030)	625	2,412
Size quintile indicators	−0.010 (0.023)	−0.053* (0.029)	625	2,412
Quadratic, total assets at g − 3	−0.008 (0.025)	−0.059* (0.032)	592	2,396
Scored firms ≤ 99th percentile of controls	−0.006 (0.024)	−0.061** (0.031)	586	2,412
Scored firms ≤ 95th percentile of controls	−0.021 (0.026)	−0.096*** (0.035)	487	2,412
Scored firms ≤ 90th percentile of controls	−0.013 (0.028)	−0.093** (0.039)	420	2,407
Doubly robust, ≤ 95th percentile	−0.010 (0.027)	−0.064* (0.034)	487	2,412
Doubly robust, ≤ 90th percentile	0.005 (0.029)	−0.061 (0.038)	420	2,407

Note: Outcome regression unless doubly robust, with firm-bootstrap standard errors; firm counts for ln(MTB). *, **, ***: 10%, 5%, 1%.

Market capitalization is more sensitive: its estimate is 0.016 without covariates, between −0.059 and −0.033 across the functional forms that adjust for size, and between −0.096 and −0.061 on the restricted supports, as expected when size enters both selection and the outcome. Table 6 describes the scored firms that the 95th percentile restriction keeps and drops. The 138 excluded firms are large (median total assets of 2,351.7 million U.S. dollars against 170.9 million) and more often Singaporean (29.0% against 1.2%), Indonesian (18.1% against 2.9%), or Thai (29.0% against 24.0%), while the retained firms are mostly Malaysian (71.5%). The restricted estimate thus describes mid-sized scored firms, largely from the Malaysian expansion.

Table 6. Scored firms retained and excluded by the 95th percentile size restriction

Characteristic	Retained	Excluded	Never-scored (all years)
Firms	487	138	2,613
Median total assets (USD million)	170.9	2,351.7	68.7
Median market capitalization (USD million)	129.5	1,478.0	42.2
Median market-to-book ratio	1.34	1.14	0.95
Mean book leverage	0.202	0.330	0.241
Mean initial ESG score	33.4	41.7	
Share Malaysian	0.71	0.15	
Share Thai	0.24	0.29	
Share Indonesian	0.03	0.18	
Share Singaporean	0.01	0.29	
Share Philippine	0.00	0.09	
Share first recorded by 2020	0.12	0.59	

Note: Scored firms in the ln(MTB) estimate, measured in base year g − 1; never-scored firms over all firm-years.

5.7 Missing Valuation Data and Recording Errors

Missing MTB ratios are not random. Firm-years without one are more often highly levered (book leverage above 0.5 in 25.4% of never-scored firm-years without an MTB ratio, against 8.4% with one) and more often Indonesian (36.2% against 24.1%; Table S8 of the supplementary material). Only 36.7% of never-scored and 36.6% of scored firm-years without an MTB ratio have a market capitalization, so most gaps are missing market data, not negative book equity alone. Among the 50 scored firms that do not enter the MTB estimate, those with base-year total assets have a median similar to that of firms that do (280.3 against 294.3 million U.S. dollars).

Table 7 checks whether these gaps and the recording errors change the estimates. Estimating log market capitalization only on firm-years that also have an MTB ratio moves its change from −0.055 to −0.023, so part of the negative market capitalization estimate comes from firm-years that drop out of the MTB sample. Keeping only firms with a complete MTB series gives 0.008 for MTB, and keeping only scored firms with complete series gives 0.006. Removing the flagged firm-years gives −0.007 for MTB and −0.045 for market capitalization; removing every firm with a flagged year, using the lower threshold of twice total assets, or winsorizing within each market and year gives MTB estimates between −0.019 and 0.005. None of these checks produces an MTB gain.

Table 7. Missing valuation data and recording errors

Specification	ln(MTB)	ln(market cap)	Scored firms in averaged event years	Control firms
Baseline	0.002 (0.024)	−0.055* (0.030)	625	2,412
Market capitalization on firm-years with an MTB ratio		−0.023 (0.029)	624	2,409
Complete MTB series, scored and control firms	0.008 (0.025)		598	1,152
Complete MTB series, scored firms only	0.006 (0.024)		598	2,412
Flagged firm-years removed	−0.007 (0.024)	−0.045 (0.029)	615	2,408
Firms with any flagged year removed	−0.008 (0.024)	−0.045 (0.029)	613	2,385
Firm-years with implied book equity above twice total assets removed	−0.019 (0.023)	−0.039 (0.028)	597	2,393
Winsorized within market and year	0.005 (0.024)	−0.064** (0.031)	625	2,412

Note: Outcome regression with firm-bootstrap standard errors; flags mark implied book equity above ten times total assets. *, **, ***: 10%, 5%, 1%.

5.8 Other Robustness Checks

None of the pre-specified rows R1 to R4 shows an MTB effect that differs from zero at the 10% level (estimates from −0.001 to 0.031), while the market capitalization and total assets results are less stable. Excluding one market at a time (row R9 and Table S9) gives MTB effects from −0.007 to 0.017. The placebo in row R5 yields 0.059 on log MTB ($p = 0.010$), restating the pre-coverage growth in Figure 1. The static TWFE regression in row R6 attributes an increase of 0.064 in log MTB ($p = 0.014$) to coverage; comparing each cohort with its own base year accounts for about half of the gap to the headline estimate, and conditioning on base-year size for most of the rest (Table S5).

5.9 Does the Initial Score Matter?

The last three rows of Table 3 divide scored firms at the cohort median of their first recorded ESG score. The difference between high- and low-score entrants is −0.041 (standard error 0.040, $p = 0.307$). Because the score rises with firm size (correlation 0.35 with base-year log total assets), Table S4 repeats the split after purging the score of size, country, and cohort; the difference is then −0.027 ($p = 0.491$). Neither split shows the sign pattern that the preference channel predicts, but with minimum detectable differences of 0.113 and 0.111, about 12% in the

MTB ratio, neither can detect a moderate difference, so the result does not show that ESG preferences are unimportant.

5.10 What the Pooled Estimate Represents

The pooled estimate weights cohorts and event years by the number of scored firms in each cell. Malaysian firms carry 55.2% of the weight in the headline MTB estimate and Thai firms 26.7%, together 81.9%; the Malaysian wave of 2021 and 2022 alone carries 44.1%, and firm-years from 2021 to 2024 carry 87.2% (Table S10). The pooled estimate therefore describes mainly the Malaysian expansion and the Thai cohorts after the COVID-19 pandemic, not an average effect for the five markets; the estimates for Indonesia, the Philippines, and Singapore in Table S11 rest on too few scored firms to support conclusions. Under the parallel trends assumption, the largest increase compatible with the headline interval is of the same order as the announcement return of 4.86% that Demiroglu and Ryngaert (2010) report for the first analyst coverage of neglected stocks, and because annual ratios miss short-lived price pressure, our estimates concern lasting revaluation (Chen et al., 2004).

6. Discussion

6.1 What the Ordering of Events Shows

The clearest result is an ordering of events: firms entered the scored universe after their MTB ratio and market capitalization had grown faster than those of never-scored firms, and their MTB ratio stopped gaining afterward. The provider may have added firms once they were large and visible enough to matter to its clients, or firms may have become scoreable by reporting, joining an index, or issuing equity; the fall in leverage toward the base year fits equity issuance. Either way, entry marks the end of a period of relative growth, not a random draw, and this finding does not rest on the parallel trends assumption or on the dating of publication.

6.2 What the Evidence Says About a Valuation Effect of Coverage

The flat post-coverage MTB path admits three readings, depending on the counterfactual trend (Rambachan & Roth, 2023; Roth, 2022). With parallel paths from the base year, the data show no detected effect of coverage (95% interval −0.046 to 0.048 in log MTB). With a continued run-up, coverage lowered valuations relative to the trend. With a partly transitory run-

up, the flat path is consistent with a positive coverage effect that offset a reversal. The comparison of datings fits a reversal, but a negative effect confined to the waves with a run-up would look the same. With annual data, a provider that chooses whom to score, and recorded scores that may precede publication, a valuation effect of coverage cannot be reliably identified without additional assumptions. One of many exploratory subgroups, the 58 firms first scored by 2019, shows 0.094 ($p = 0.047$), which we do not read as evidence of a gain because it is not adjusted for multiple comparisons.

6.3 Mechanisms That the Data Can and Cannot Evaluate

Recognition predicts a lasting gain for every newly scored firm; we detect none under parallel trends, but without ownership, trading, or fund-holding data we cannot tell whether recognition did not increase or whether it increased without moving prices. Preferences predict opposite signs for high- and low-score entrants, and the split in Section 5.9 cannot detect a moderate difference. The absence of a detected gain fits the evidence on certification by sustainability indices (Durand et al., 2019; Hawn et al., 2018) and contrasts with the price effects of MSCI downgrades (Berg, Heeb, et al., 2022). Two features of our setting may explain the contrast: U.S. ESG fund holdings respond to MSCI ratings, not to those of other providers (Berg, Heeb, et al., 2022), and scored firms were large and visible before coverage, which leaves less room for recognition (Merton, 1987). Unlike U.S. evidence that coverage changes firm conduct and payout (Bikmetova & Pirinsky, 2026; Tsang et al., 2024, 2025), we find no stable change in leverage or total assets. The run-up also fits evidence that realized green returns reflect unexpected shifts in ESG demand (Pástor et al., 2022), but without ownership data we cannot tell whether ESG investors bought these firms before or after coverage.

6.4 Implications

For researchers, comparisons of scored and unscored firms, static TWFE estimates, and backtests built on the first recorded score can attribute selection to the score (Berg et al., 2020; Sahin et al., 2023); studies of ESG ratings in Southeast Asia should model when firms enter coverage, report pre-entry dynamics, and record the download date of their rating data. For rating providers and data users, the scored universe in these markets tilts toward large firms and toward firms that had outgrown their peers, and publishing the date on which each firm's coverage

began, with the vintages of revised scores, would separate what was known from what was filled in later.

For investors, a portfolio screened on ESG scores in these markets holds firms that, on average, entered coverage after outgrowing their peers, so the screen tilts toward recent winners and away from smaller firms, and the date a firm became screenable is the publication date, which may follow the recorded year. For exchanges, regulators, and listed firms, broad expansions such as the Malaysian wave appear to select less on prior growth, and we detect no MTB gain within it, although its interval (−0.053 to 0.091) admits gains larger than the pooled upper bound. Our results do not show that expanding ESG coverage has no valuation benefit; they show that such a benefit cannot be established from these data, so arguments for expanding coverage should rest on grounds other than an expected rise in issuer valuations.

7. Limitations

The study has eight limitations. First, the provider chooses when to cover a firm, and the post-coverage estimates rest on conditional parallel trends that the pre-coverage estimates question; our sensitivity analysis uses a linear trend and a simplified bound, not the confidence sets of Rambachan and Roth (2023). Second, we observe neither publication dates nor announcement returns. Third, we observe only LSEG coverage, so for firms with earlier FTSE Russell or other scores the event is an additional score. Fourth, the data contain no returns, ownership, or index membership. Fifth, MTB ratios are missing for 18.5% of never-scored and 9.0% of scored listed firm-years, more often for distressed and Indonesian firms, and some market capitalization records contain scale errors. Sixth, Malaysian and Thai firms make up 84.3% of scored firms. Seventh, the sample includes only firms that survived to 2024. Eighth, with an MDE of 0.067 on log MTB, smaller effects would be detected with less than 80% probability.

8. Conclusion

Entry into LSEG ESG coverage in five Southeast Asian markets was selective. The firms that received a first recorded score were larger than never-scored firms, and on average their market-to-book ratio had grown 10.0% more than that of similar firms from five years to one

year before entry. The run-up was largest in the Thai and Singaporean wave of 2019 and 2020, present in the remaining cohorts, and not detected in the broad Malaysian wave of 2021 and 2022 (95% interval −7.4% to 15.2%). After entry, the valuation gap stopped widening. Whether coverage itself changed valuations cannot be settled with these data: the answer moves from no detected change (95% interval −0.046 to 0.048 in log MTB) to a decline of 6.6% depending on when investors are assumed to have seen the score, the decline appears only where valuations had run up, and modest departures from parallel trends change either reading.

Research and investment screens that start from a provider's universe inherit the provider's selection. Point-in-time records of when each firm's score was first published, and ownership and trading data, would allow future work to test whether ESG coverage in Southeast Asian markets changes who holds a firm and at what price.

References

Avramov, D., Cheng, S., Lioui, A., & Tarelli, A. (2022). Sustainable investing with ESG rating uncertainty. *Journal of Financial Economics*, *145*(2), 642–664. https://doi.org/10.1016/j.jfineco.2021.09.009

Baker, A. C., Larcker, D. F., & Wang, C. C. Y. (2022). How much should we trust staggered difference-in-differences estimates? *Journal of Financial Economics*, *144*(2), 370–395. https://doi.org/10.1016/j.jfineco.2022.01.004

Berg, F., Fabisik, K., & Sautner, Z. (2020). *Is history repeating itself? The (un)predictable past of ESG ratings* (Finance Working Paper No. 708/2020). European Corporate Governance Institute. https://ssrn.com/abstract=3722087

Berg, F., Heeb, F., & Kölbel, J. F. (2022). *The economic impact of ESG ratings* (SAFE Working Paper No. 439). Leibniz Institute for Financial Research SAFE. https://ssrn.com/abstract=4088545

Berg, F., Kölbel, J. F., & Rigobon, R. (2022). Aggregate confusion: The divergence of ESG ratings. *Review of Finance*, *26*(6), 1315–1344. https://doi.org/10.1093/rof/rfac033

Bernama. (2022, November 2). *Bursa Malaysia signs sustainability MoU with London Stock Exchange Group*. https://www.bernama.com/en/news.php?id=2134522

Bikmetova, N., & Pirinsky, C. A. (2026). Do ESG rating agencies improve ESG performance? *Journal of Business Ethics*, *204*(2), 335–365. https://doi.org/10.1007/s10551-025-06063-0

Bursa Malaysia. (2015). *Sustainability reporting guide*. Bursa Malaysia Berhad.

Callaway, B., & Sant'Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, *225*(2), 200–230. https://doi.org/10.1016/j.jeconom.2020.12.001

Chen, H., Noronha, G., & Singal, V. (2004). The price response to S&P 500 index additions and deletions: Evidence of asymmetry and a new explanation. *The Journal of Finance, 59*(4), 1901–1929. https://doi.org/10.1111/j.1540-6261.2004.00683.x

Christensen, D. M., Serafeim, G., & Sikochi, A. (2022). Why is corporate virtue in the eye of the beholder? The case of ESG ratings. *The Accounting Review*, *97*(1), 147–175. https://doi.org/10.2308/TAR-2019-0506

Crump, R. K., Hotz, V. J., Imbens, G. W., & Mitnik, O. A. (2009). Dealing with limited overlap in estimation of average treatment effects. *Biometrika*, *96*(1), 187–199. https://doi.org/10.1093/biomet/asn055

Demiroglu, C., & Ryngaert, M. D. (2010). The first analyst coverage of neglected stocks. *Financial Management*, *39*(2), 555–584. https://doi.org/10.1111/j.1755-053X.2010.01084.x

Dobrick, J., Klein, C., & Zwergel, B. (2023). Size bias in Refinitiv ESG data. *Finance Research Letters*, *55*, Article 104014. https://doi.org/10.1016/j.frl.2023.104014

Drempetic, S., Klein, C., & Zwergel, B. (2020). The influence of firm size on the ESG score: Corporate sustainability ratings under review. *Journal of Business Ethics*, *167*(2), 333–360. https://doi.org/10.1007/s10551-019-04164-1

Durand, R., Paugam, L., & Stolowy, H. (2019). Do investors actually value sustainability indices? Replication, development, and new evidence on CSR visibility. *Strategic Management Journal*, *40*(9), 1471–1490. https://doi.org/10.1002/smj.3035

Gibson Brandon, R., Krueger, P., & Schmidt, P. S. (2021). ESG rating disagreement and stock returns. *Financial Analysts Journal*, *77*(4), 104–127. https://doi.org/10.1080/0015198X.2021.1963186

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, *225*(2), 254–277. https://doi.org/10.1016/j.jeconom.2021.03.014

Harris, L., & Gurel, E. (1986). Price and volume effects associated with changes in the S&P 500 list: New evidence for the existence of price pressures. *The Journal of Finance*, *41*(4), 815–829. https://doi.org/10.1111/j.1540-6261.1986.tb04550.x

Hartzmark, S. M., & Sussman, A. B. (2019). Do investors value sustainability? A natural experiment examining ranking and fund flows. *The Journal of Finance*, *74*(6), 2789–2837. https://doi.org/10.1111/jofi.12841

Hawn, O., Chatterji, A. K., & Mitchell, W. (2018). Do investors actually value sustainability? New evidence from investor reactions to the Dow Jones Sustainability Index (DJSI). *Strategic Management Journal*, *39*(4), 949–976. https://doi.org/10.1002/smj.2752

Kelly, B., & Ljungqvist, A. (2012). Testing asymmetric-information asset pricing models. *The Review of Financial Studies*, *25*(5), 1366–1413. https://doi.org/10.1093/rfs/hhr134

Krueger, P., Sautner, Z., Tang, D. Y., & Zhong, R. (2024). The effects of mandatory ESG disclosure around the world. *Journal of Accounting Research*, *62*(5), 1795–1847. https://doi.org/10.1111/1475-679X.12548

LSEG. (2025). *LSEG ESG scores (ESG score field), annual, 2014–2024* [Data set]. LSEG Workspace.

LSEG. (2023). *Refinitiv Data Platform ESG bulk point in time user guide* (Document version 2). https://developers.lseg.com/content/dam/devportal/api-families/refinitiv-data-platform/refinitiv-data-platform-apis/documentation/rdp_esg_pit_user_guide.pdf

Merton, R. C. (1987). A simple model of capital market equilibrium with incomplete information. *The Journal of Finance*, *42*(3), 483–510. https://doi.org/10.1111/j.1540-6261.1987.tb04565.x

Nguyen, T. T. (2025). Ownership structure and corporate social responsibility: Evidence from Vietnam. *Southeast Asian Journal of Economics*, *13*(2), 67–98.

Otoritas Jasa Keuangan. (2017). *Peraturan Otoritas Jasa Keuangan Nomor 51/POJK.03/2017 tentang penerapan keuangan berkelanjutan bagi lembaga jasa keuangan, emiten, dan perusahaan publik* [Regulation on the application of sustainable finance to financial services institutions, issuers, and public companies]. https://www.ojk.go.id/sustainable-finance/id/peraturan/peraturan-ojk/Documents/SAL%20POJK%2051%20-%20keuangan%20berkelanjutan.pdf

Pástor, Ľ., Stambaugh, R. F., & Taylor, L. A. (2021). Sustainable investing in equilibrium. *Journal of Financial Economics*, *142*(2), 550–571. https://doi.org/10.1016/j.jfineco.2020.12.011

Pástor, Ľ., Stambaugh, R. F., & Taylor, L. A. (2022). Dissecting green returns. *Journal of Financial Economics*, *146*(2), 403–424. https://doi.org/10.1016/j.jfineco.2022.07.007

Pedersen, L. H., Fitzgibbons, S., & Pomorski, L. (2021). Responsible investing: The ESG-efficient frontier. *Journal of Financial Economics, 142*(2), 572–597. https://doi.org/10.1016/j.jfineco.2020.11.001

Prukumpai, S., Sethapramote, Y., & Luangaram, P. (2022). Political uncertainty and the Thai stock market. *Southeast Asian Journal of Economics*, *10*(3), 227–257.

Rambachan, A., & Roth, J. (2023). A more credible approach to parallel trends. *The Review of Economic Studies*, *90*(5), 2555–2591. https://doi.org/10.1093/restud/rdad018

Roth, J. (2022). Pretest with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights*, *4*(3), 305–322. https://doi.org/10.1257/aeri.20210236

Sahin, Ö., Bax, K., Paterlini, S., & Czado, C. (2023). The pitfalls of (non-definitive) environmental, social, and governance scoring methodology. *Global Finance Journal*, *56*, Article 100780. https://doi.org/10.1016/j.gfj.2022.100780

Sant'Anna, P. H. C., & Zhao, J. (2020). Doubly robust difference-in-differences estimators. *Journal of Econometrics*, *219*(1), 101–122. https://doi.org/10.1016/j.jeconom.2020.06.003

Securities and Exchange Commission Philippines. (2019). *Memorandum Circular No. 4, series of 2019: Sustainability reporting guidelines for publicly-listed companies*. https://www.sec.gov.ph/wp-content/uploads/2020/09/2019MCNo4.pdf

Shanaev, S., & Ghimire, B. (2022). When ESG meets AAA: The effect of ESG rating changes on stock returns. *Finance Research Letters*, *46*(Part A), Article 102302. https://doi.org/10.1016/j.frl.2021.102302

Shleifer, A. (1986). Do demand curves for stocks slope down? *The Journal of Finance*, *41*(3), 579–590. https://doi.org/10.1111/j.1540-6261.1986.tb04518.x

Singapore Exchange. (2016). *SGX Mainboard Rules 711A and 711B: Sustainability report* [Listing rules]. SGX Rulebook. https://rulebook.sgx.com/rulebook/711a

S&P Global Market Intelligence. (2025). *Compustat Global, annual fundamentals, 2014–2024* [Data set].

Sudsawasd, S., & Sakdapat, P. (2026). Transaction costs and stock market behavior: Evidence from Thailand. *Southeast Asian Journal of Economics*, *14*(1), 257–287.

Tilleke & Gibbins. (2020, September). *Thailand's SEC unifies annual report for listed companies*. https://www.tilleke.com/insights/thailands-sec-unifies-annual-report-listed-companies/

Tsang, A., Wang, Y., Xiang, Y., & Yu, L. (2024). The rise of ESG rating agencies and management of corporate ESG violations. *Journal of Banking & Finance*, *169*, Article 107312. https://doi.org/10.1016/j.jbankfin.2024.107312

Tsang, A., Wang, Y., Xiang, Y., & Yu, L. (2025). ESG ratings and dividend changes: Evidence from the initiation of nonfinancial agency coverage. *Corporate Governance: An International Review*, *33*(4), 554–577. https://doi.org/10.1111/corg.12615